

Exam Corrector

*An offline computer-vision platform that grades
bubble-sheet exams from an ordinary printed
page and a phone photo — no scanner, no cloud,
no proprietary forms.*

PREPARED BY	Amirsam Eftekharinia
STACK	Python · OpenCV · Flask
DEPLOYMENT	Local, offline
TEST SUITE	52 tests, passing
LANGUAGES	English · Persian (RTL)

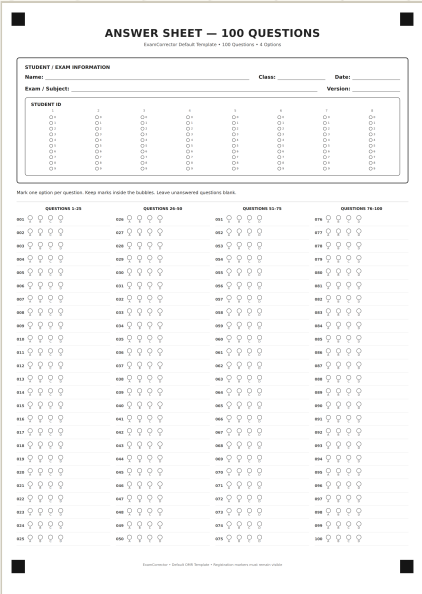


FIG. 1 — DEFAULT 100-QUESTION ANSWER SHEET, AS PRINTED

Grading multiple-choice exams by hand doesn't scale, and the machines that replace it come with strings attached.

A class of thirty students sitting a 100-question exam produces three thousand answers to mark. Done by hand, that's slow and quietly error-prone — tired eyes miscount rows, transcription slips happen between the answer key and the gradebook. The commercial fix, Scantron-style OMR hardware, solves the speed problem but adds new ones: dedicated scanners, proprietary answer sheets, and often a recurring cloud subscription just to read a piece of paper.

Treat the exam sheet as the interface, not the hardware.

ExamCorrector reads a plain answer sheet — printed on any home or school printer — from any camera or scanner image: a phone photo, a flatbed scan, a screenshot. All processing happens locally; nothing leaves the machine it runs on. Registration relies on four printed corner markers rather than a specialized scanner bed, so a page photographed at a slight angle, under uneven light, or even inside a wider screenshot still registers correctly.

0.00px

mean corner-marker reprojection error after registration, measured across the test fixtures

93.3%

bubble-reading confidence retained under a simulated 50-unit directional lighting gradient

100×4

questions and options on the default template — configurable per exam

Why this matters in practice: a photo taken by hand is never perfectly lit or perfectly flat. Most of the engineering effort in this project went into making the reading robust to that reality, rather than assuming a clean scanner-bed image.

Who this is for

Built with the classroom teacher in mind first — someone grading their own exams between other duties, without an IT department or a purchasing budget behind them. The same offline, no-account, no-recurring-cost shape suits any setting where a laptop and a printer are available but a specialized scanner or a cloud subscription is not: tutoring centers, community programs, and schools in low-connectivity areas among them.

Five stages, each one built to fail honestly rather than guess.

A running principle across the whole pipeline: when the system isn't sure, it says so — flagged for a human to check — rather than silently reporting a confident-looking wrong answer.

STAGE 1	STAGE 2	STAGE 3	STAGE 4	STAGE 5
Registration	Bubble reading	Identity	Grading	Review
Detects the four corner markers and computes a homography, correcting rotation, skew, and camera-angle distortion.	Four independent darkness measures vote on each bubble; measured relative to its own local background, not a fixed brightness constant.	Student-ID bubbles and handwritten Name/Class fields are read and, where possible, cross-checked against an uploaded roster.	Deterministic scoring against the answer key, with configurable rules for blank and multiple-marked responses.	Low-confidence answers are queued for human review; every correction is kept alongside the original reading, never overwriting it.

The core robustness mechanism

Each bubble's fill is judged against the paper immediately surrounding it, not a fixed ink/paper brightness assumed in advance. A shadow falling across one side of the page darkens a bubble and its neighboring paper by roughly the same amount, so the *relative* reading stays accurate even when the *absolute* brightness has shifted. This one design decision is what makes the 93.3%-under-shadow figure on the previous page possible.

Four votes per bubble

No single darkness measure is trusted alone. Fill ratio, an Otsu-threshold ratio, an adaptive-threshold ratio, and edge density are computed independently and required to agree before a mark counts as confident — disagreement among them is itself a signal, routed to human review rather than resolved by picking a winner.

Erasures, not just marks

A partially erased bubble doesn't look like a clean mark or clean blank — it looks uneven. The interior-brightness variance of each bubble is checked specifically for that unevenness, so a changed answer is caught rather than misread as the original one.

Reading student identity without inventing it

Handwriting OCR is treated as best-effort, not authoritative: a field the system can't read confidently is reported as unreadable rather than filled in with a guess. A bubble-based Student ID, when read at full confidence, can be cross-matched against an uploaded class roster to attach a verified name — more reliable than trusting handwriting alone.

Built to be checked, including checking its own checks.

Every change to the grading logic runs against an automated suite before it ships. More importantly, the project's own testing method has itself been audited — twice a self-consistent test was found to be passing for the wrong reason, and both times the fix was a test that checks reality independently, not just internal agreement.

52

automated integration and unit tests, covering registration, bubble consensus, shadow robustness, Student-ID geometry, and OCR field extraction — run after every change, currently all passing.

A concrete example: catching a self-validating bug

An early Student-ID layout was verified by placing synthetic test marks and reading them back with the same coordinate formula used to grade real sheets — every test passed at 100% confidence. On a real photographed sheet, the same layout misread several digits. The two methods agreed with each other while both being wrong relative to the printed page: a self-consistent test, not a true one.

The fix was a verification method that never trusts the code under test: independently detecting the physical bubble positions in the rendered image and checking the stored coordinates against *that*, instead of against themselves. That check is now a permanent, automated regression test, run against every shipped answer-sheet template.

FIX	Bubble radius miscalibration — template stored a fixed constant instead of the sheet's actual measured bubble size	caught via shadow-robustness testing
FIX	Student-ID grid coordinates offset from the physical page by ~15px	caught via independent ground-truth verification
FIX	Name/Class OCR silently failing when the field's OS-level text engine was missing	now surfaced with install guidance, not a silent blank
FIX	Wide, mostly-blank text-field crops confusing OCR segmentation	fields now trimmed to actual ink before reading

The full lifecycle of an exam, not just the scan step.

Exam workflow

- Answer-key entry with a keyboard-driven editor
- Batch upload — images or multi-page PDF
- Review queue for anything flagged low-confidence
- Manual correction for answers and identity, original reading always kept alongside the fix
- Class roster import with ID-based cross-matching

Output & interface

- Export to CSV, multi-sheet Excel, and PDF
- Per-question analytics across a class
- Color-coded diagnostic view of every sheet
- Full English / Persian interface, right-to-left layout included
- Light and dark themes

Ships with a ready-to-print 100-question, 4-option answer sheet in English and Persian. New layouts can be designed and calibrated from a single photo of a blank page — the system measures the grid geometry automatically rather than requiring it hand-coded per template.

What's solid, and what's still in front of it.

DONE	Core CV pipeline: registration, bubble consensus, shadow robustness	52/52 tests passing
DONE	Full exam workflow, roster matching, multi-format export	verified end-to-end
DONE	Bilingual interface with full RTL support	English & Persian
OPEN	One-click desktop packaging (single executable)	build scripts exist, unverified on this environment
OPEN	Background/asynchronous batch processing	currently synchronous; fine at classroom scale
OPEN	Per-student individual report cards	exam-level reporting only, today
OPEN	Fully general custom-template calibration	non-standard Student-ID layouts need manual verification

Design principle carried through the whole project: when the system is uncertain, it reports that uncertainty rather than resolving it silently in either direction. The same standard applies to this document — every number in it was measured, not estimated.



A grading tool built the way the exams it reads deserve to be graded — carefully, and without guessing.

ExamCorrector is an active, working project: the pipeline, the test suite, and every figure in this document run today, offline, from a normal printed page.

CONTACT

Amirsam Eftekharinia

STATUS

Active development
